



End Term (Even) Semester Examination, May 2026

Roll no. 2519216

Name of the Course: B.Tech.

Semester: II

Name of the Paper: Engineering Mathematics II

Paper Code: TMA 201

Time: 3:00 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b and c.
- (iii) Total marks of each main question are 20. Each sub-question of a main question carries 10 marks.

Q1.

(10 × 2=20 Marks)

a. Find the complete solution for the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x \sin x$. (CO1)

b. Solve by the method of variation of parameters. (CO1)

$$\frac{d^2y}{dx^2} + 4y = \tan 2x$$

c. Solve the homogeneous linear differential equation. (CO1)

$$x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x$$

Q2.

(10 × 2=20 Marks)

a. Solve the following differential equation by series solution method. (CO2)

$$(1 + x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$$

b. Prove that $xJ'_n = -nJ_n + xJ_{n-1}$. (CO2)

c. Show that $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$ (CO2)

Q3

(10 × 2=20 Marks)

a. Test the convergence of the series. (CO3)

$$1 + \frac{3}{5}x^2 + \frac{8}{10}x^3 + \frac{15}{17}x^4 + \dots + \frac{n^2 - 1}{n^2 + 1}x^n + \dots$$

b. Test whether the following series is convergent or not (CO3)

$$\sum \left\{ \left(\frac{n+1}{n} \right)^{n+1} - \left(\frac{n+1}{n} \right) \right\}^{-n}$$

c. Find the interval of convergence for the following power series. (CO3)

$$\sum_{n=0}^{\infty} \frac{2^n}{n^2} (x-3)^n$$



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Q.4

(10 × 2 = 20 Marks)

- a. Show that the following function satisfies CR-equations at $z=0$, but not analytic at that point.

(CO4)

$$f(z) = \begin{cases} \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2}, & z \neq 0 \\ 0, & 0 \end{cases}$$

- b. Find the bilinear transformation that maps the points $z = 0, -1, i$ into the points $w = i, 0, \infty$. Hence, find the image of $|z| = 1$.

(CO4)

- c. If $u = x^2 - y^2$, find the corresponding analytic function $f(z) = u + iv$ using Milne-Thompson's method.

(CO4)

Q.5

(10 × 2 = 20 marks)

- a. Evaluate the following integral using the Cauchy integral formula.

(CO5)

$$\int_C \frac{4-3z}{z(z-1)(z-2)} dz, \quad C: |z| = \frac{3}{2}.$$

- b. Evaluate the following integral using the Cauchy residue theorem.

(CO5)

$$\oint_{|z|=2} \frac{e^z}{z^2(z-1)} dz$$

- c. Obtain the Laurent's series expansion of $f(z) = \frac{1}{(z-1)(z-2)}$ for $1 < |z| < 2$.

(CO5)